

Renewable Energy Engineering

Solar Energy – Basics and Solar Geometry

Introduction

- The **sun** is a large sphere of intensely hot gaseous matter ($\approx 75\%$ hydrogen and 25% helium), continuously generating the energy by thermonuclear fusion reaction (in which hydrogen atoms are converted in to helium atoms). This energy is radiated from the sun in all directions and a very small fraction of it reaches the earth.

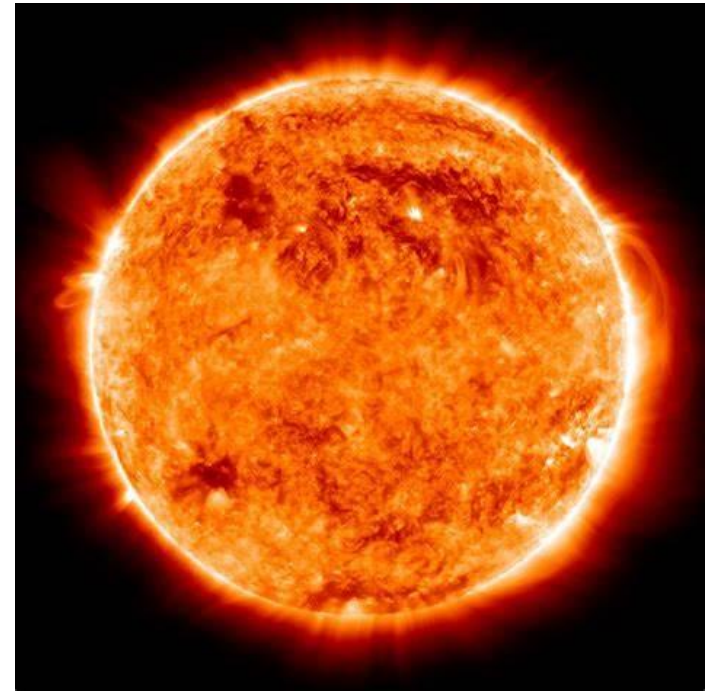
- The principal characteristics of sun are,

$$\text{Mass, } M = (1.991 \pm 0.002) \times 10^{30} \text{ kg}$$

$$\text{Radius, } R = (6.960 \pm 0.001) \times 10^8 \text{ m}$$

$$\text{Average Density, } \rho = (1.410 \pm 0.002) \times 10^3 \text{ kg/m}^3$$

$$\text{Temperature (Average on the surface), } T = 5762 \pm 50 \text{ K}$$



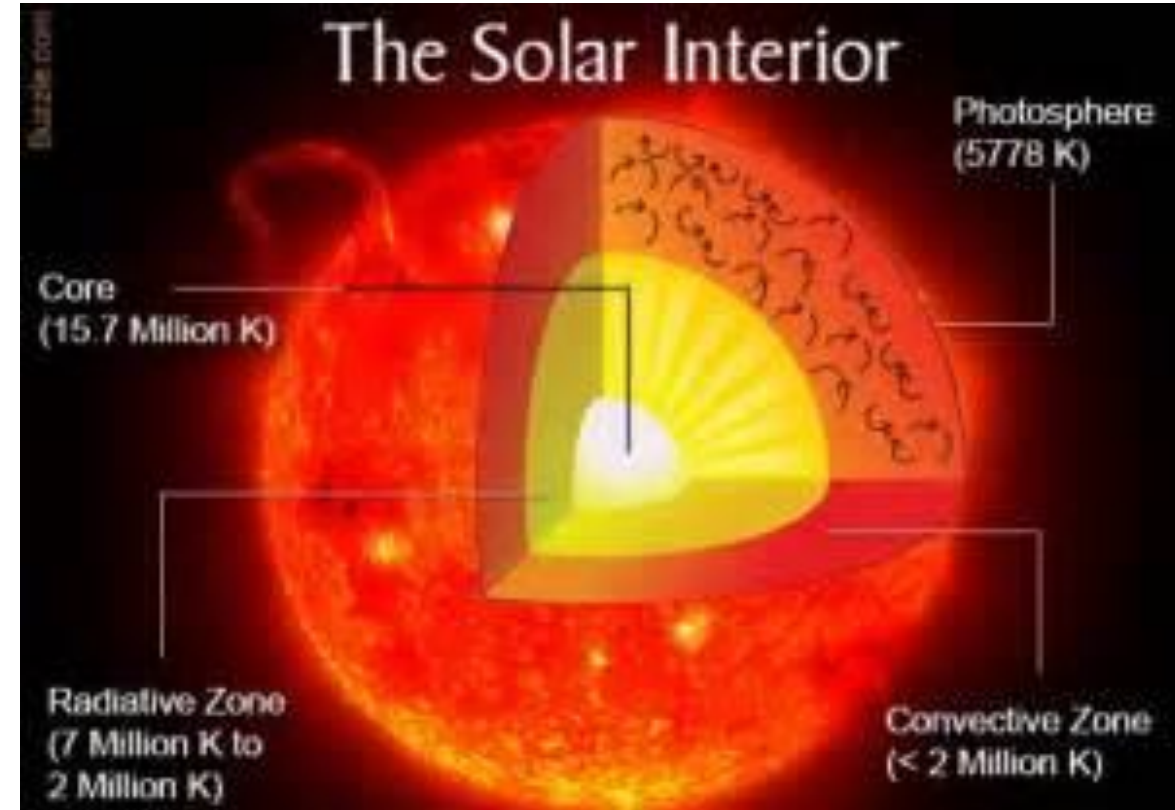
- As seen from the earth, sun rotates about its axis once in every 4 week, but not as a rigid body.
- The equator takes ≈ 27 earth days and polar region takes ≈ 30 earth days for each rotation.

Structure of Sun

➤ The structure of the sun is generally divided into following regions:

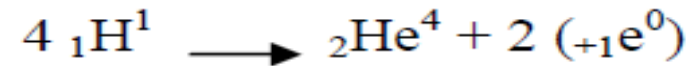
- Solar interior or Core
- Radiative Zone
- Convective Zone
- Solar atmosphere

➤ **Solar interior** or **Core** constitute the main mass of the sun and has gases at a pressure of 250 billion atmosphere, temperature of 8×10^6 to 40×10^6 K and density of 80 to 100 times that of water.



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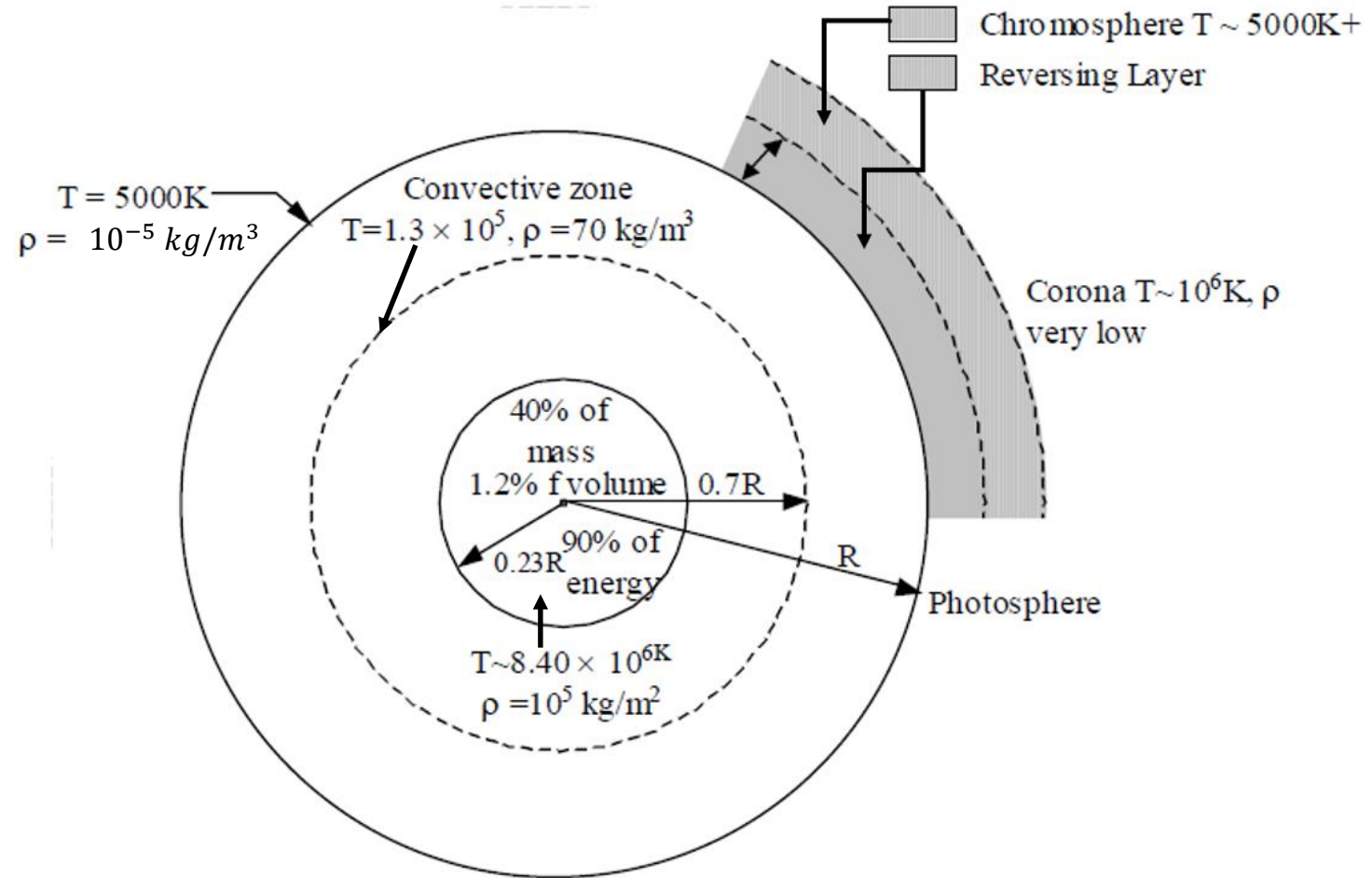
- Solar interior act as continuous fusion reactor with its constituent gases as the “containing vessel” retained by gravitational forces.
- The energy generated in solar interior because of continuous conversion of 4 atoms of hydrogen in to one nucleus of helium and two positrons in series of reaction, involving other particles such as carbon, nitrogen, other nuclei etc. that are continuously appearing and disappearing in reaction.



- The mass of helium nucleus is lesser than that of the hydrogen (proton) mass. Thus, the mass is lost in the reaction.
- The lost mass is converted into energy as per law $E = mc^2$, which maintains the temperature of core several millions of degree.
- The energy produced in solar interior is transferred to the surface and radiated into space.
- 90% of the energy is generated in region 0 to 0.23R (R = radius of sun), which contains 40% mass of sun.

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- The energy generated in solar interior is transferred outward by radiation in the form of X and γ -rays up to the distance of $\approx 0.7R$ from the centre, the zone is known as **Radiative Zone**, where the temperature dropped to $\approx 1.3 \times 10^5$ K and density dropped to $\approx 70 \text{ kg/m}^3$.
- The region of sun in between $0.7R$ to $1.0R$, where the fluid is the medium and the transfer of energy is mainly by convection, is known as **Convective Zone**.

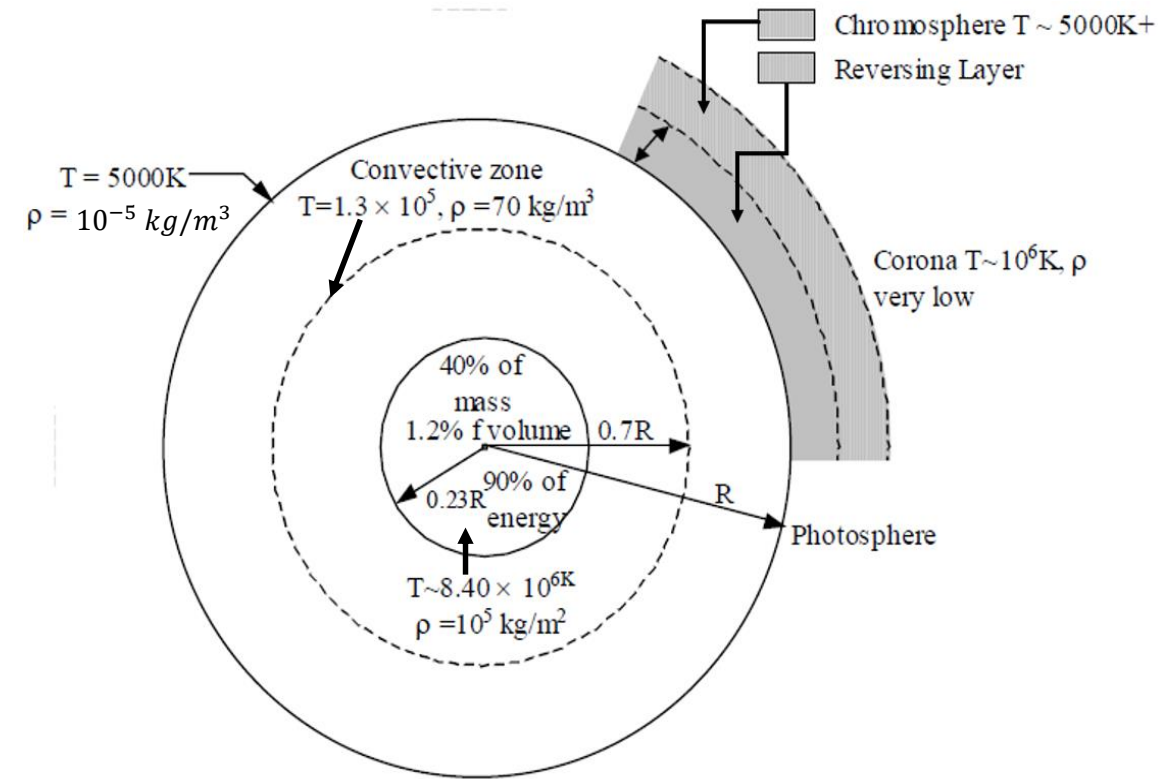


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- In convective zone, the density dropped to 10^{-5} kg/m^3 and temperature falls to $\approx 5000 \text{ K}$.
- The outer layer of convective zone is called as **Photosphere**, which is the source of light and heat.
- Photosphere is essentially opaque, as the gases of which it is composed are strongly ionized and able to absorb and emit a continuous spectrum of radiation.
- Outside the photosphere is a more or less transparent solar atmosphere, observable during total solar eclipse or by instruments that occult (cannot explained by science) the solar disk.
- Above the photosphere, a layer of cooler gases several hundred kilometers deep called the **Reversing layer**.
- Outside of that, a transparent layer of 10000 km thick consist of rarefied gas (pressure below atmospheric pressure) is called as **Chromosphere**.

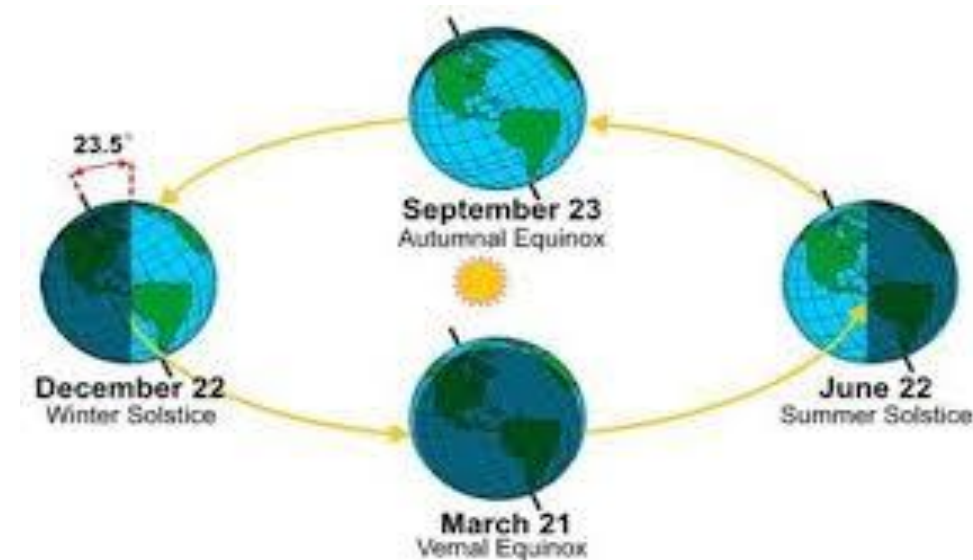
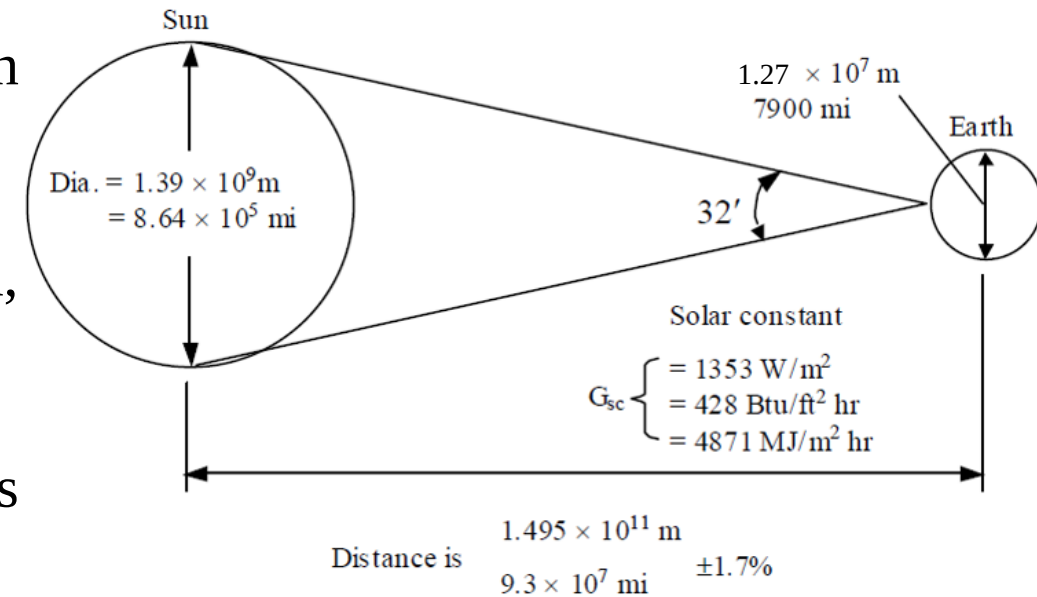
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- Chromosphere is gaseous layer with temperature higher than the photosphere and has lower density.
- The red colour of chromosphere is resulting from the strong α line of hydrogen in the spectrum of layer; a continuous band of low intensity mainly caused by the scattering of solar radiation.
- Temperature of the chromosphere is $\approx 5000\text{ K}$ at base and 10^6 K at elevation of 12000 kms.
- Finally, the **Corona** - a whitish glowing layer and its beauty may be observed during total solar eclipse.
- **Corona** consists of highly ionized gases of very low density and temperature of $\approx 10^6\text{ K}$.



Sun-Earth Size and Position

- The geometry of the sun-earth relationships is shown in Figure.
- The diameter of the sun is 1.39×10^9 m and that of earth, 1.27×10^7 .
- The mean earth distance is 1.495×10^{11} m. The sun subtends an angle of $32'$ with the earth.
- The distance between the sun and earth varies by 1.7 % due to the elliptic orbit of the earth around the sun.
- The distance between the earth and the sun determines the intensity of solar radiation that can be received by the earth.
- The elliptic motion of the earth around the sun causes the seasonal variation (shown in figure) in the solar radiation received by the earth.



Solar Constant

- The solar radiation received from the sun on earth is almost parallel because of the large distance between sun and earth.
- Measurement indicates that the solar energy flux received from the sun on **outside the earth's atmosphere** (known as **extra-terrestrial**) is essentially **constant**.
- The **solar constant** (I_{SC}) is the rate at which energy is received from the sun on a unit area perpendicular to the rays of sun, at the mean distance of the earth from the sun.
- The recently reported value of the **solar constant** is **1367 W/m²**.
- Because of the elliptical motion of earth around the sun, the distance between the earth and sun varies throughout the year that cause the variation in extra-terrestrial solar flux.

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- The radiation incident on surface varies inversely with the square of the earth-sun distance.
- The value of **solar flux** on **extra-terrestrial surface** on particular day can be calculated using equation:

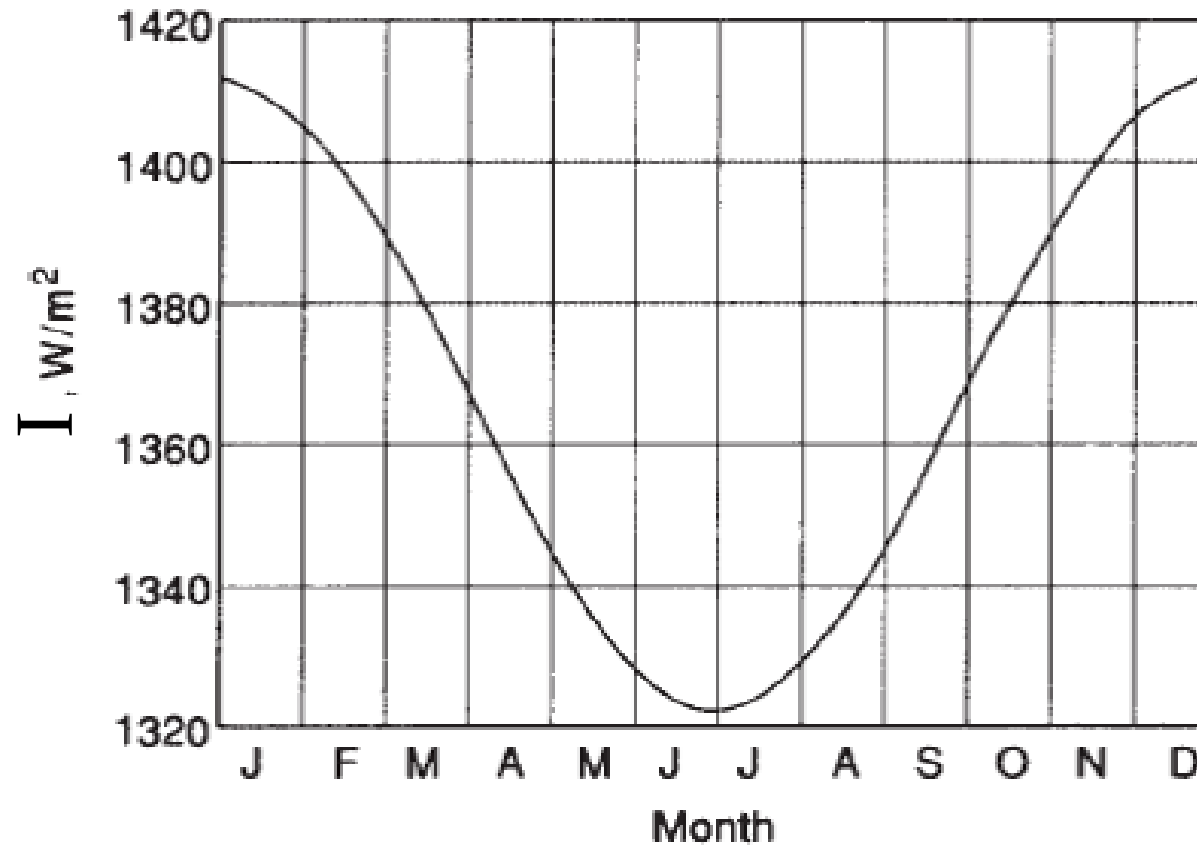
$$\frac{I}{I_{sc}} = 1 + 0.033 \cos \left(\frac{360 \times n}{365} \right)$$

where n is day of the year, I is the intensity of solar radiation on extra-terrestrial surface on particular day of the year.

- e.g. for 13th April, $n = 31(\text{January}) + 28(\text{February}) + 31(\text{March}) + 13(\text{April}) = 113$
- The cosine function varies from +1 to -1, the extra-terrestrial radiation flux varies by $\pm 3.3\%$.

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- Variation in extra-terrestrial solar radiation flux throughout the year,



- Energy emitted by sun $\approx 3.8 \times 10^{26} \text{ W}$
- Energy received on extra-terrestrial surface of earth $\approx 1.7 \times 10^{18} \text{ W}$

Electromagnetic Energy Spectrum

➤ The spectrum of electromagnetic radiation is divided into wavelength band as:

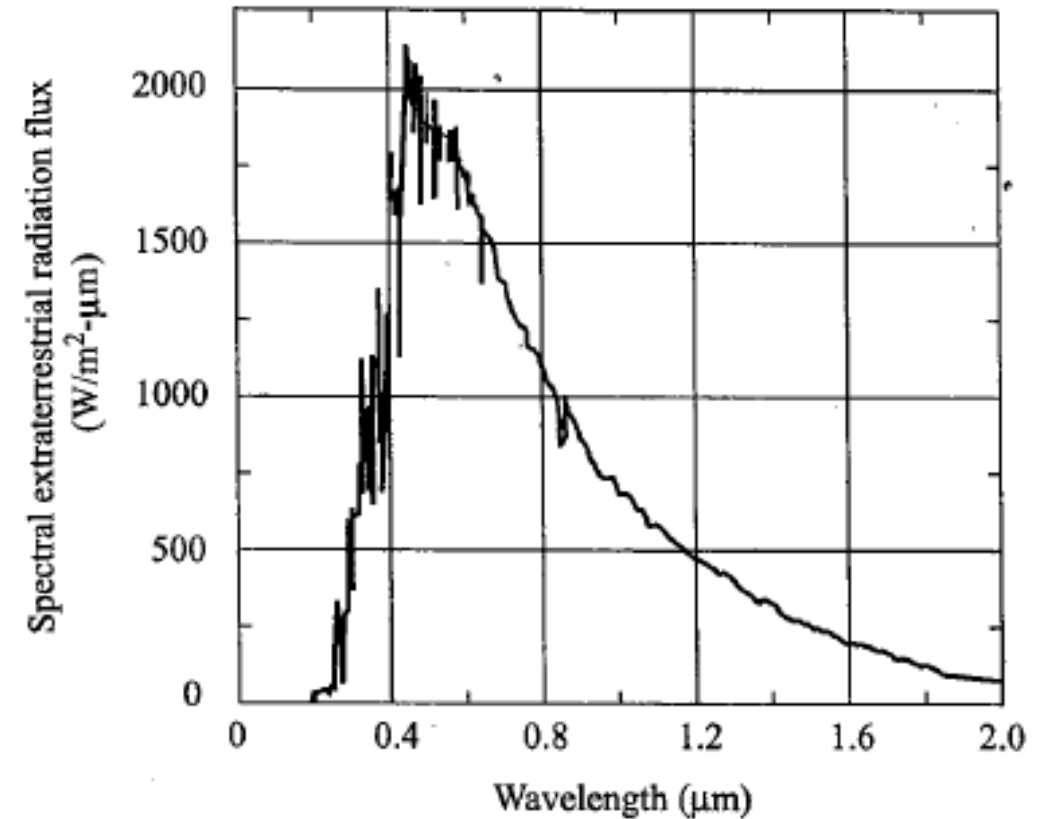
Wave length range (Approximately)	Name
$\lambda < 10^{-8} \mu$	Cosmic rays
$10^{-8} \mu < \lambda < 10^{-4} \mu$	Gamma rays
$10^{-5} \mu < \lambda < 10^{-2} \mu$	X- rays
$2 \times 10^{-2} \mu < \lambda < 0.38 \mu$	Ultra-violet rays
$0.38 \mu < \lambda < 0.78 \mu$	Visible light rays
$0.78 \mu < \lambda < 10^2 \mu$	Infra-red rays
$10^2 \mu < \lambda < 10^{10} \mu$	Radio Waves

➤ From the whole spectrum of electromagnetic radiation, human can sense only those rays that have wavelength between ≈ 0.1 to 100μ , which heats up the body of human is called the “**Thermal Radiation**”.

Spectral Distribution of Solar Radiation

- Spectral distribution of extra-terrestrial solar radiation is shown in figure.
- The area under the curve is solar radiation.

Wavelength range (μ)	0 - 0.38	0.38 – 0.78	0.78 – ∞
Approximate energy (W/m^2)	87	656	623
Approximate % of total energy	6.4	48.0	45.6



- The visible light rays of solar energy spectrum carries $\approx$ half of the total energy.

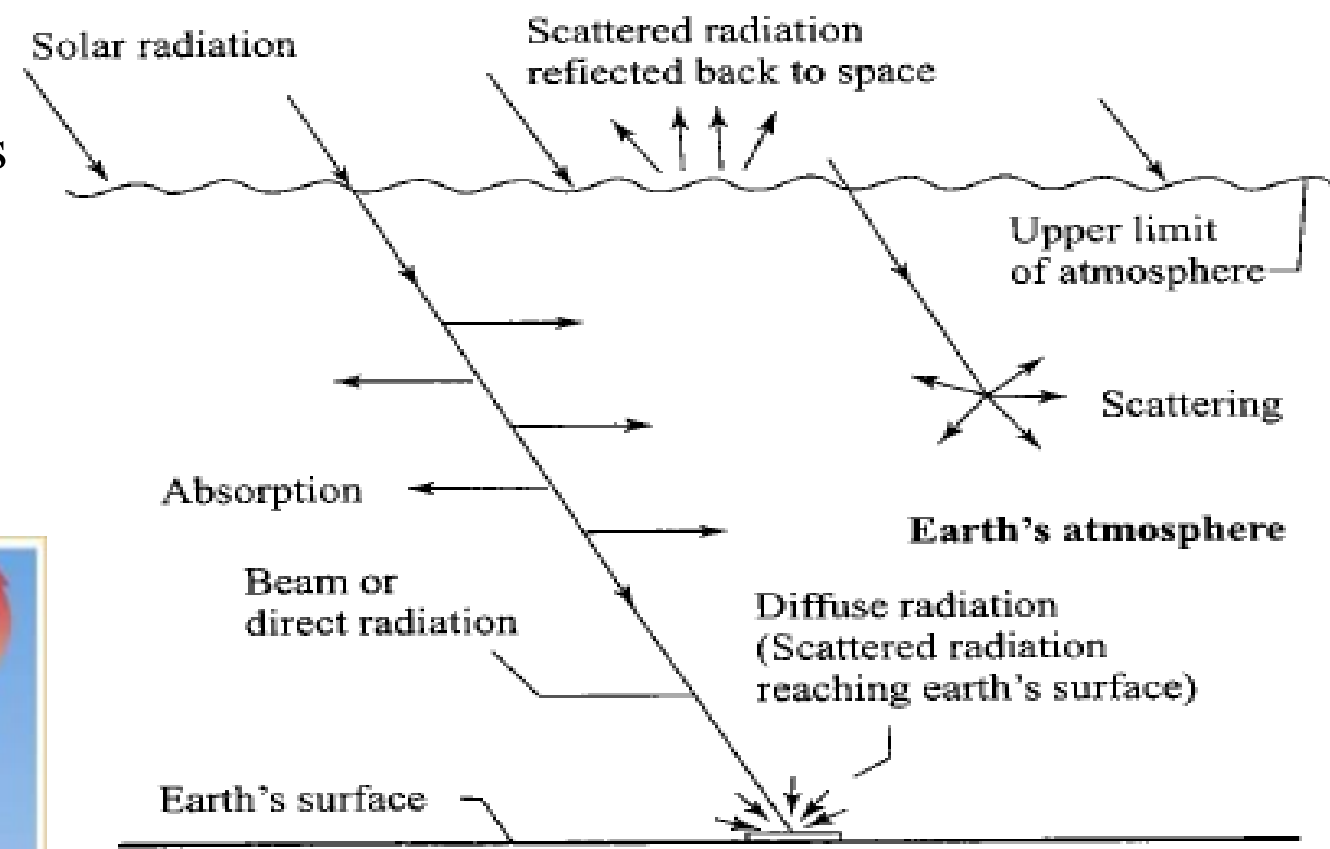
Solar radiation at the Earth's Surface or Terrestrial surface

- Solar radiation received at the terrestrial surface of earth in an attenuated (reduced) form because of the mechanisms of **absorption** and **scattering** as solar radiation passes through the earth's atmosphere.
- As the solar radiation passes through earth's atmosphere, **Absorption of short-wave length ultraviolet rays** occurs because of the presence of **Ozone** and **long wave length infra red rays** by **carbon dioxide and moisture** present in atmosphere.
- The absorption of solar radiation increases the internal energy of atmosphere and hence increases the temperature of atmosphere.
- **Scattering** occurs due to the gaseous molecules, dust particles and particulate molecules available in atmosphere.
- The scattered radiation is redistributed in all the direction, some going back into space and some reaching to earth's surface.
- The solar radiation received on terrestrial surface of earth is in the range of $0.29\ \mu$ to $2.5\ \mu$.

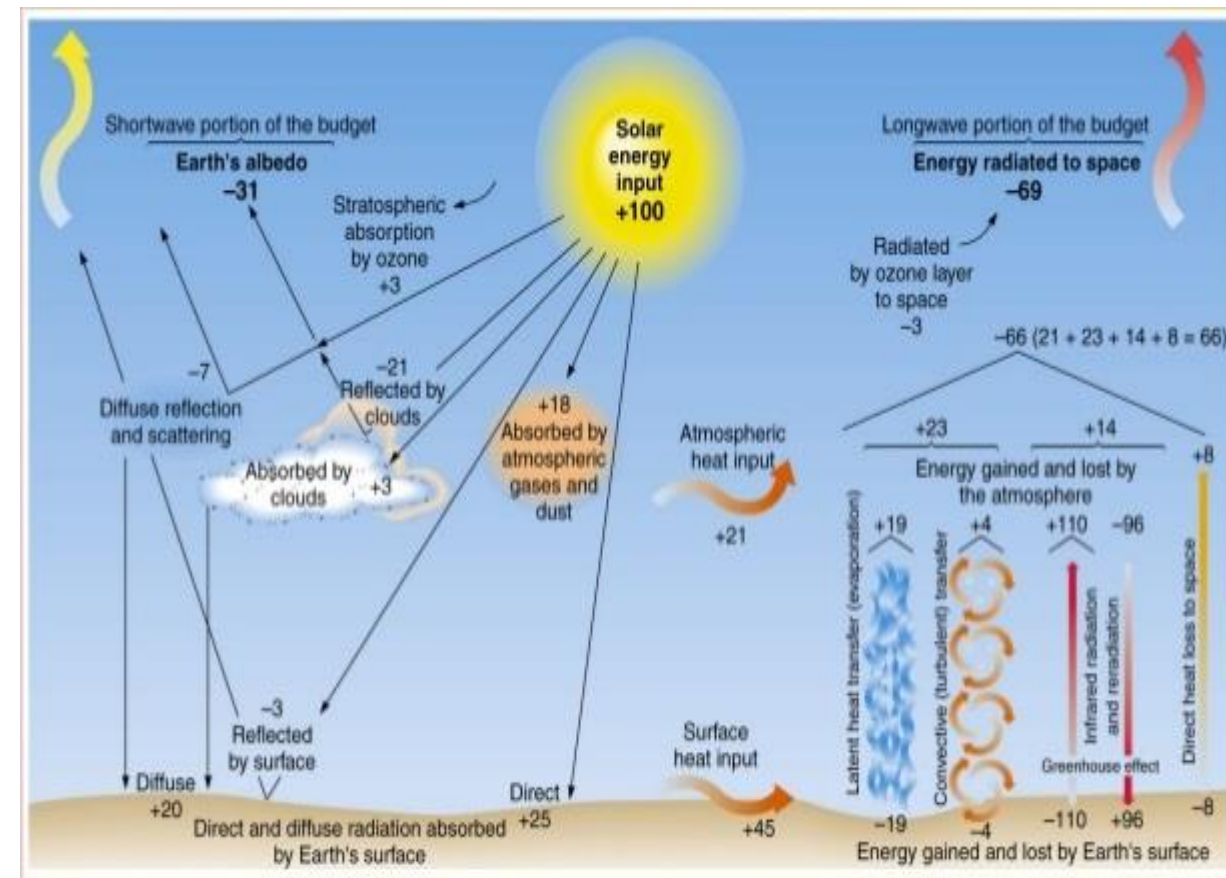
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➤ The atmosphere at any location on the earth's surface is classified into:

- Atmosphere without clouds
- Atmosphere with clouds



- The mechanisms of absorption and scattering are similar for both the cases.
- However, less attenuation takes place in a cloudless sky and maximum solar radiation is received on earth's surface.



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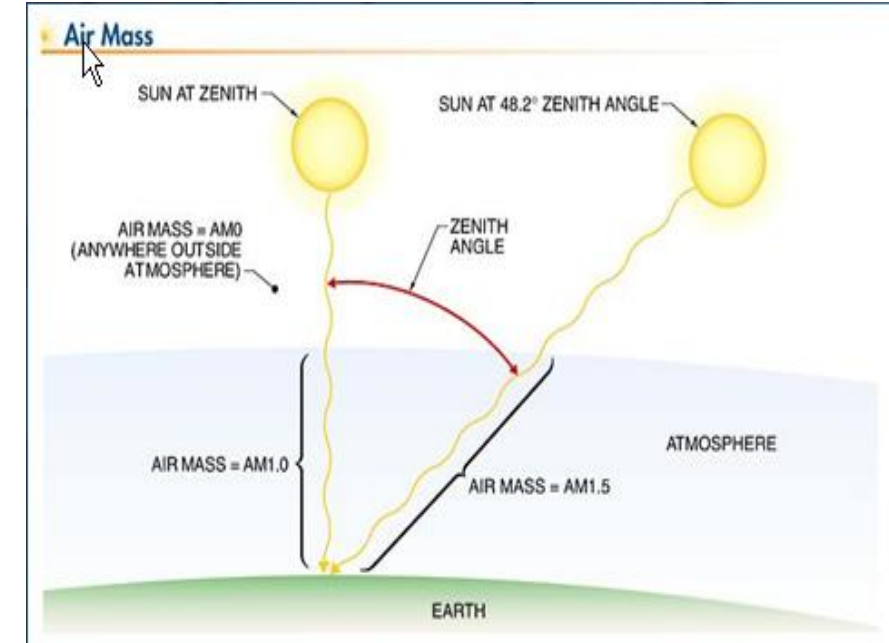
- The solar radiation received at the earth's surface **without change of direction** *i.e.* in line with the location of sun or as originate from sun is called **Beam or Direct radiation**.
- The **solar radiation received on extra-terrestrial surface** is **entirely a beam or direct radiation**.
- The solar radiation received at the earth's surface **with change of direction** *i.e.* from all parts of the sky's hemisphere after scattering in atmosphere is called **Diffuse radiation**.

Beam radiation + Diffuse radiation = Total or Global radiation

- The intensity of diffuse radiation coming from the various direction in the sky is not uniform and said to be anisotropic (different value when measured in different direction) in nature.
- In part or heavy cloud situation, diffuse radiation is modelled as perfectly uniform and said to be isotropic in nature.

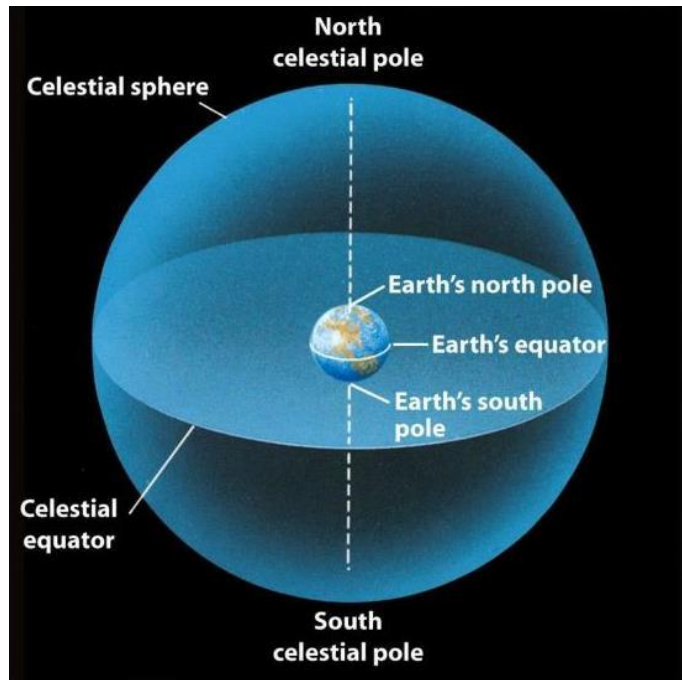
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- **Sun at Zenith:** the position of sun directly overhead of observer.
- **Zenith angle (θ_z):** is the angle made by the sun's ray or beam radiation with normal to a horizontal surface.
- **Air mass** is used as a measure of the distance travelled by beam radiation through the atmosphere before it reaches a location on the earth's surface or terrestrial surface.
- **Air mass (m) :** is the ratio of mass of the atmosphere through which the beam radiation passes to the mass of atmosphere through which beam radiation passes when sun is at zenith or directly overhead of observer.
 - $m = \sec\theta_z$, for zenith angle from 0° to 70°
 - $m = 1$ when sun is at zenith (AM1)
 - $m = 2$ for zenith angle of 60° (AM2)
 - $m = 0$ for extra-terrestrial surface (AM0)

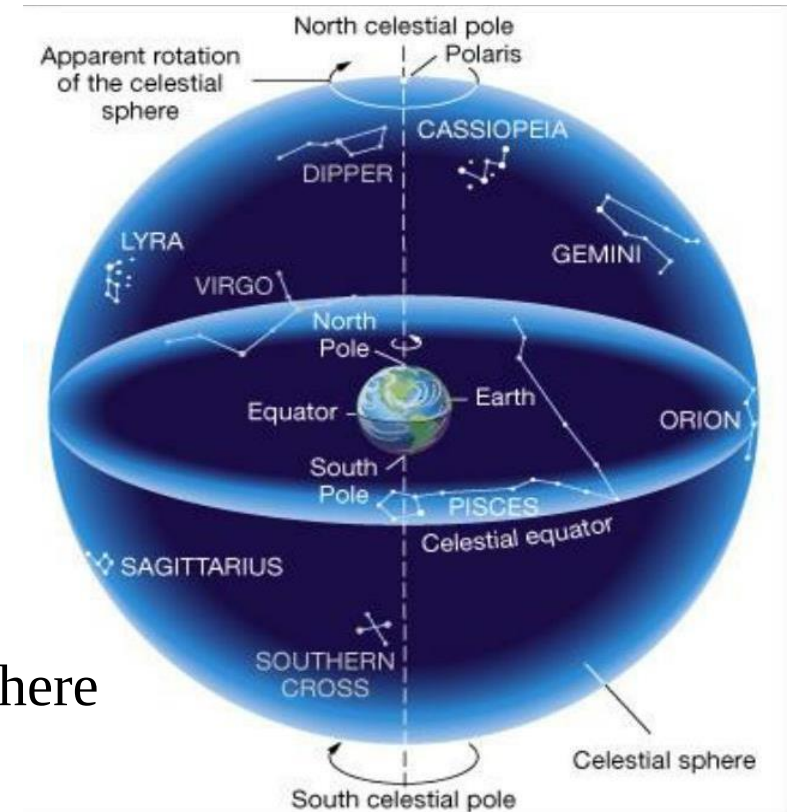


Celestial Vault

- **Celestial Sphere:** is an imaginary sphere of arbitrarily large radius concentric with earth and assumed that all the objects (stars, planets, moon etc.) in the observer's sky can be thought of as projected upon inside surface (*i.e.* located at same distance away from observed) of celestial sphere.

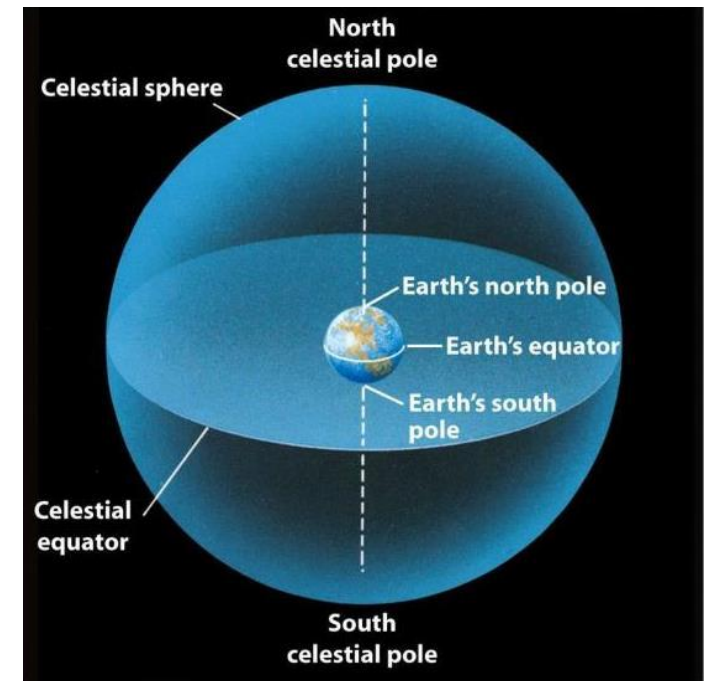
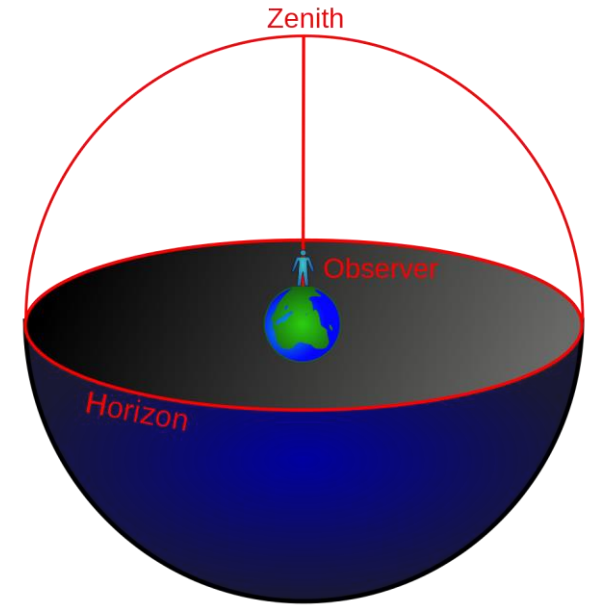


- **Zenith:** is the point on celestial sphere directly over the observer's head.
 - Zenith changes with location of observer.
- **Nadir:** is the point on the celestial sphere diametrically opposite to the zenith.
 - Nadir also changes with location of observer.
 - However, Nadir is not visible like Zenith.



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- **Zenith:** is the vertical line joining the zenith to the center of earth.
- **Visible Horizon :** is the apparent location where the celestial sphere meets the ground.
- As the visible horizon is uneven. So that, observer cannot define the location in sky with respect to visible horizon.
- **Astronomical Horizon:** is the even circle (which may be either below or above the visible horizon) at a same distance away from the zenith, an angular distance of 90° .
- **Poles of earth:** the ends of the axis of rotation of the earth mark two important points on earth's surface, called the **poles of earth**.
- One as North pole and other as South pole.



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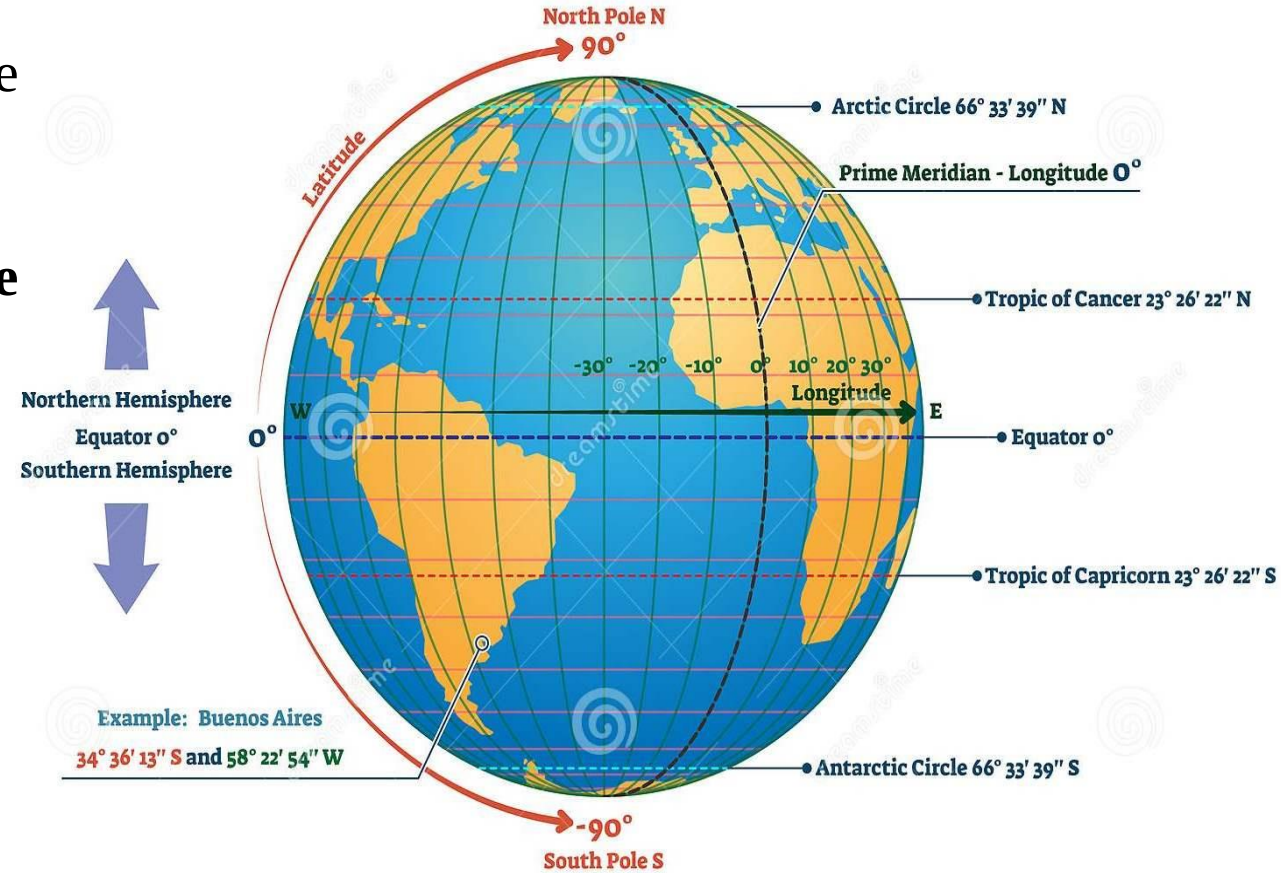
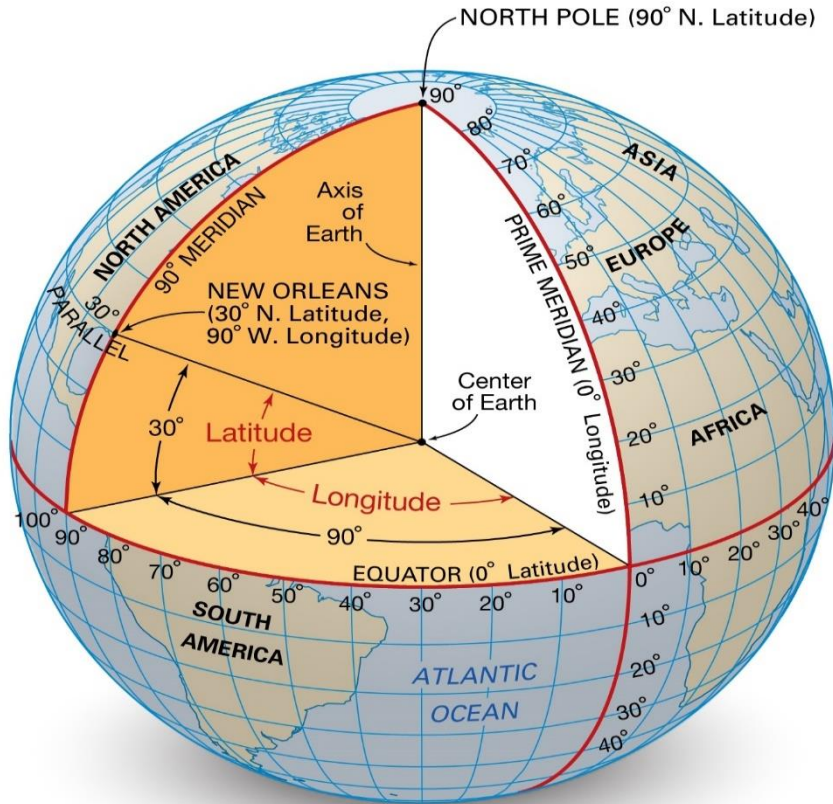
- **Earth's equator:** is an imaginary great circle normal to the earth's axis, dividing the distance between the earth's poles along its surface into two equal parts.
- Equator divides earth in to two hemisphere called **Northern hemisphere** and **Southern hemisphere**.
- **Prime Meridian:** is an imaginary great circle passing through the location of Royal Observatory Greenwich, outside of London and two poles of earth, intersecting the equator at right angle.
- Prime meridian divides earth in to two hemisphere called **Western hemisphere** and **Eastern hemisphere**.
- The great circles similar to prime meridian have been drawn at an interval of 15° through two poles of earth.
- With reference to prime meridian, position of place on earth surface is located.



Solar Geometry

➤ **Longitude (L):** of location is the angular distance measured east or west of the prime meridian.

- Longitude is **Zero (0°)** degree for **Prime meridian**.
- **0 to 180° E (+Ve), 0 to 180° W (−Ve)**

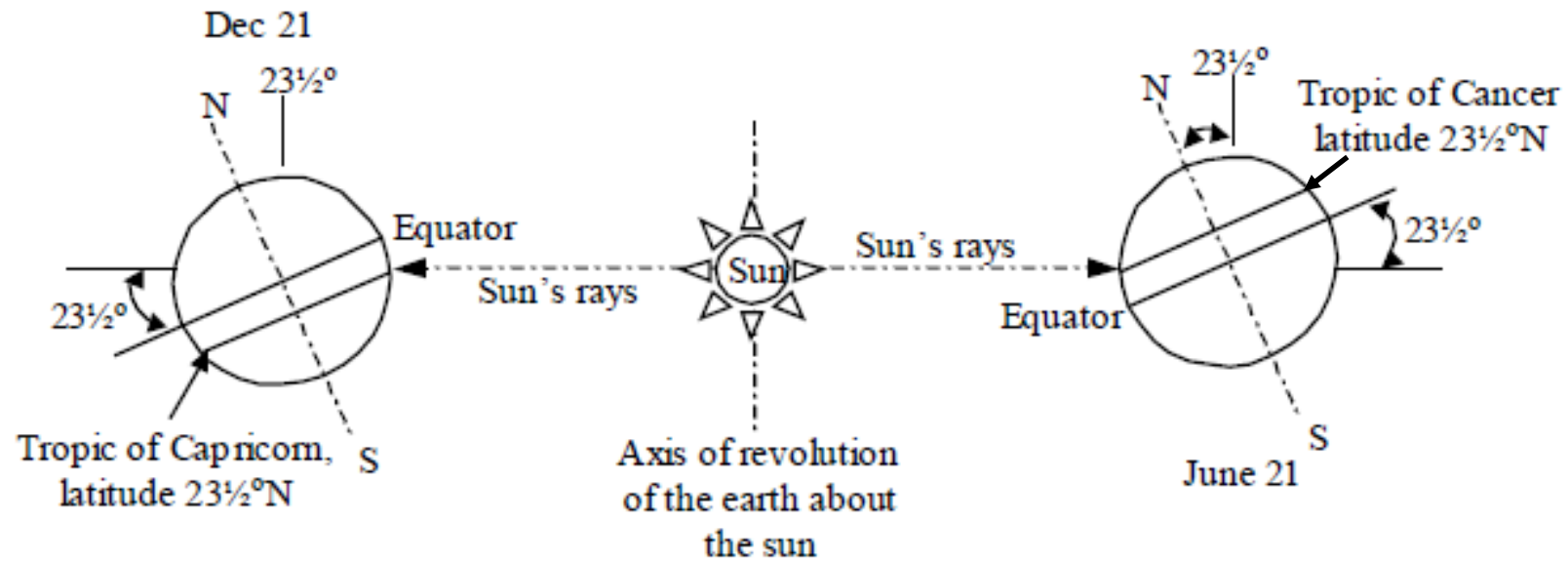
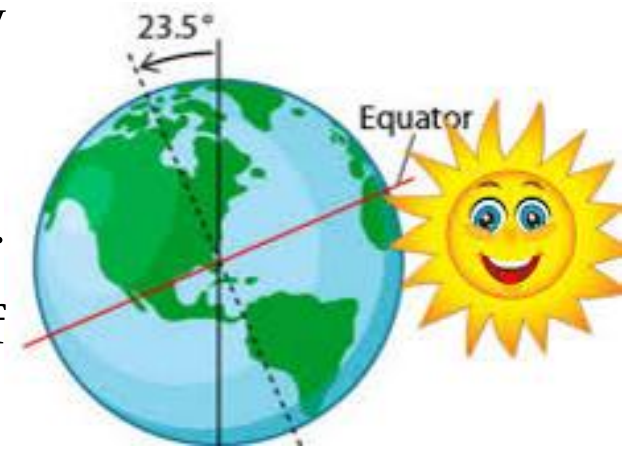


➤ **Latitude (Φ):** of location is the angular distance north or south of equator measured from the center of earth.

- Latitude is **Zero (0°)** degree for **Equator**.
- 0 to 90° N (+Ve), 0 to 90° S ($-Ve$)**

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- **Declination angle (δ , in degree):** is the angular distance of sun's ray north or south of equator.
- Declination angle vary between **23.45° on June 21** to **-23.45° on December 21**.
- At the time of winter solastice (mid winter in northern hemisphere, tropic of Capricorn), the sun rays would be 23.45° south of earth's equator ($\delta = -23.45^\circ$).
- At the time of summer solastice (mid summer in northern hemisphere, tropic of Cancer), the sun rays would be 23.45° north of earth's equator ($\delta = 23.45^\circ$).
- At the **equinox** (nearly equal day equal night, **21st March and 22nd September**), the **declination angle is Zero (0°) degree**.



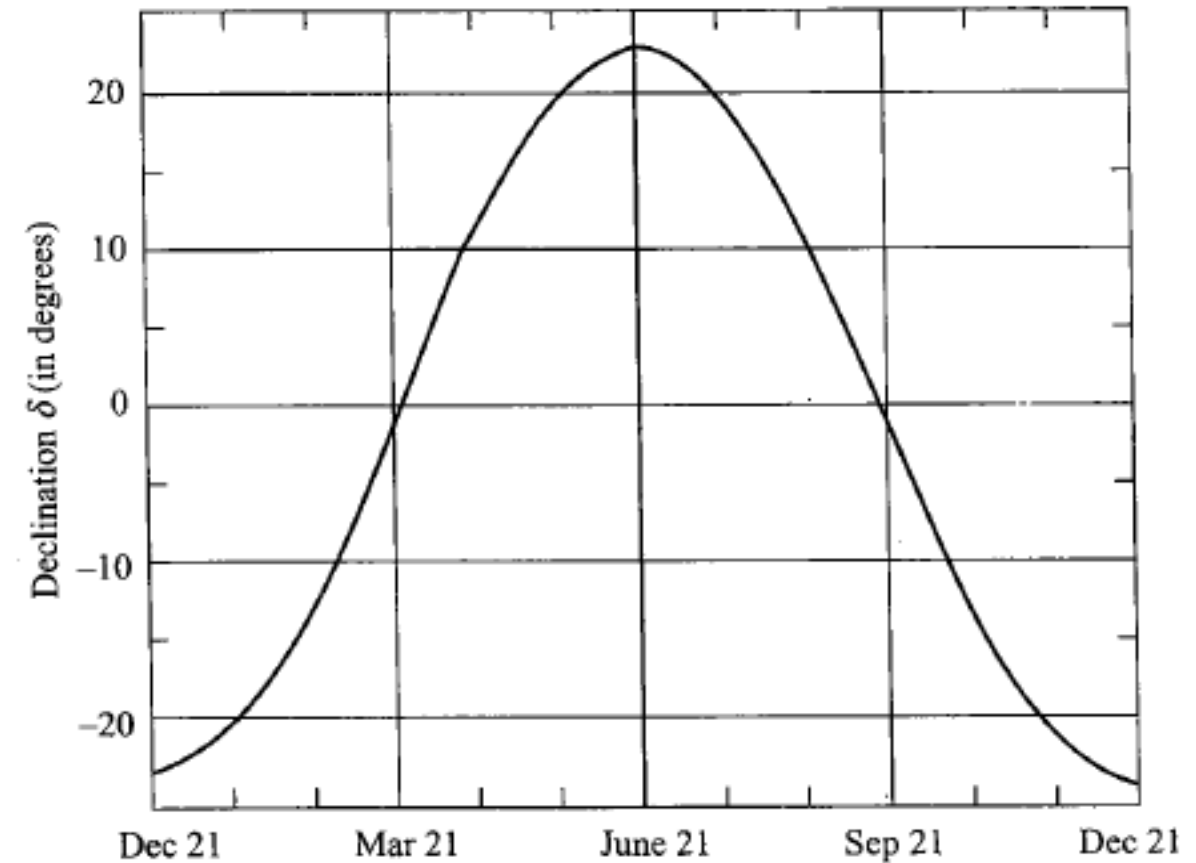
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- On **equinox day**, sun ray's or beam radiation is directly over the equator of earth at solar noon (Sun at Zenith).
- The **Declination angle (δ in degree)** on particular day of the year is calculated by cooper equation,

$$\delta = 23.45 \sin \left(\frac{360 \times (284 + n)}{365} \right)$$

where n is day of the year.

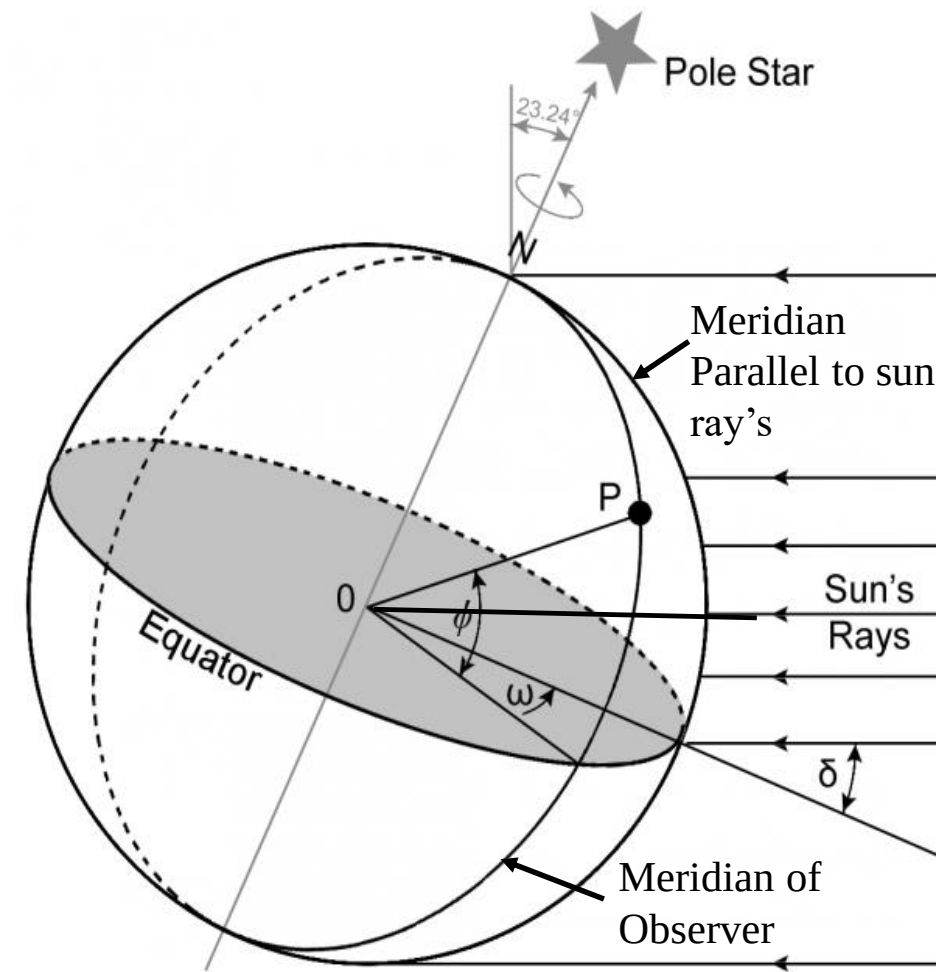
- Variation in declination angle throughout the year is,



- The **Declination angle** is reckoned as “-Ve” from 22nd September to 21st March and “+Ve” from 21st March to 22nd September.

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- **Hour angle (ω , in degree):** is the angle through which the earth must turn to bring the meridian or longitude of location directly in line with the sun's ray or beam radiation.
- In figure, angle between the line OP and projection of OP on the earth equatorial plane is **latitude of location P**.
- Angle between the line extending from center of sun to the center of earth and projection of this line upon the earth's equatorial plane is **Declination angle**.
- Angle measured in earth's equatorial plane between the projection of line OP and the projection of line extending from center of sun to the center of earth is **Hour angle**.
- **Hour angle, $\omega = 15 (\text{LST} - 12)$** in degree, where **LST** is **Local Solar Time** in hours.
- **1 Hour is equivalent to $15^\circ (= 360^\circ/24)$.**
- **Hour angle** is reckoned “-Ve” in the **forenoon** duration and “+Ve” in the **afternoon**.
- At **Solar noon (Sun at Zenith)**, hour angle is **Zero ($\omega = 0$)**.



Solar Time

- Hour angle and all the values of time in solar energy computation are expressed in terms of **Local Solar Time (LST)** or **Local Apparent Time (LAT)**.
- **LST** is the time based on the apparent angular motion of the sun across the sky, with *solar noon* being the time when sun crosses the longitude / meridian of the observer.
- Therefore, required to convert the clock time in to **LST** or **LAT**.
- The clock time is reckoned with reference to Greenwich or Prime meridian (0° longitude) and from the mid night, is known as universal time of **Greenwich Civil Time (GCT or GMT)**.
- **Local Civil Time (LCT)** is reckoned from the longitude of the place.
- LCT is precisely of 24 hours whereas LST is slightly different due to irregularities of earth rotation in elliptical orbit, increase in velocity at perihelicon (position of earth nearest to sun) etc.
- Hence, the **length of apparent solar day** (the interval between two successive passage of sun through meridian / longitude of location) is **not constant**.

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- Therefore, LST may deviate from LCT and the difference **between the LST and LCT** is called **Equation of Time (E.O.T.)**.

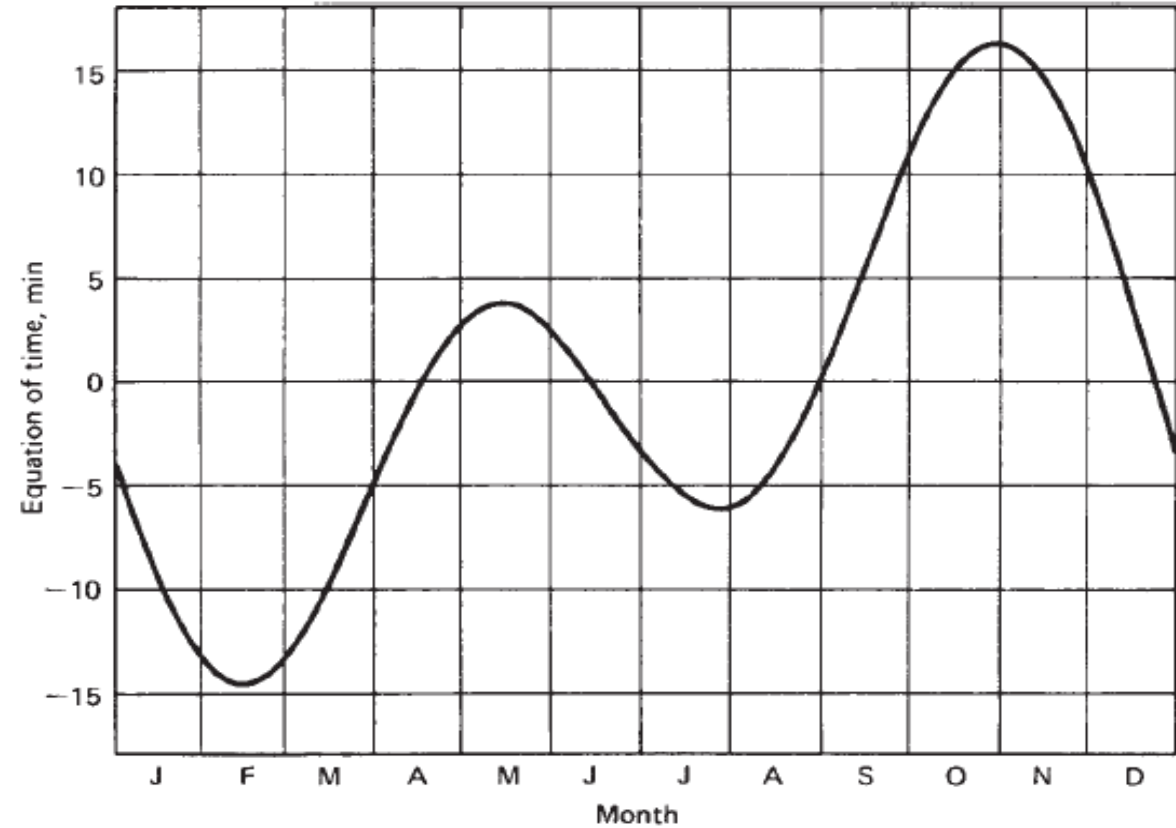
$$\text{LST} = \text{LCT} + \text{Equation of Time (E.O.T.)}$$

- **E.O.T.** is calculated using relation,

$$\text{E. O. T.} = 9.87 \sin 2B - 7.53 \cos B - 1.5 \sin B$$

where $B = \frac{360(n-81)}{364}$, n is day of the year.

- The variation in E.O.T. throughout the year is shown in figure.



- At a given location, **LCT** may differ from the **clock time**.
- Clock time is usually set for the same time throughout the entire zone covering about **15°** of meridian, known as **Standard Time (ST)**.

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- The **standard time** kept in each zone is corresponding to longitude / meridian near to the center of meridians of each zone.
- In India, standard time corresponds to longitude **82.5° E** and known as **Indian Standard Time (IST)**.
- Therefore, **LCT** can be derived from the **ST** using relation:

$$\mathbf{LCT = ST \pm 4 \times (L_{ST} - L_{Local})}$$

$$\mathbf{LST \text{ or } LAT \text{ (in minutes)} = LCT + E.O.T. = ST \pm 4 \times (L_{ST} - L_{Local}) + E.O.T.}$$

where **ST** = standard time for the zone (in minutes), **L_{ST}** = longitude for the local time zone (standard time) (in degree), **L_{Local}** = longitude of location (in degree), **E.O.T.** (in minutes), “ + ” sign for western hemisphere, “ – ” sign for eastern hemisphere, “ **4** ” indicates earth takes 4 minutes to traverse **1°** of longitude.

- For India, $\mathbf{LST \text{ or } LAT \text{ (in minutes)} = ST - 4 \times (L_{ST} - L_{Local}) + E.O.T.}$

Problem-1

- Determine the declination angle, local solar time and hour angle at 3:00 pm (IST) on June 15 for the location (**18° 54' N, 72° 49' E**).

- **Solution:**

Declination angle (in degree), $\delta = 23.45 \sin \left(\frac{360 \times (284 + n)}{365} \right)$

- $n = \text{January} + \text{February} + \text{March} + \text{April} + \text{May} + 15\text{th June}$
- $n = 31 + 28 + 31 + 30 + 31 + 15 = 166 \text{ days}$

$$\delta = 23.45 \sin \left(\frac{360 \times (284 + 166)}{365} \right) = 23.31^\circ$$

$$\text{LST (in minutes)} = \text{ST} - 4 \times (\text{L}_{\text{ST}} - \text{L}_{\text{Local}}) + \text{E.O.T.}$$

- $\text{E.O.T.} = 9.87 \sin 2B - 7.53 \cos B - 1.5 \sin B$

Cont...

- $B = \frac{360 (n-81)}{364} = \frac{360 (166-81)}{364} = \mathbf{84.07^\circ}$
- $E.O.T. = 9.87 \sin(2 \times \mathbf{84.07^\circ}) - 7.53 \cos(\mathbf{84.07^\circ}) - 1.5 \sin(\mathbf{84.07^\circ}) = \mathbf{-0.24 \text{ min}}$
- $ST = 3.00 \text{ pm} = 15 \text{ h} = 15 \times 60 = \mathbf{900 \text{ min}}$
- $L_{ST} = \mathbf{82.5^\circ}$
- $L_{Local} = 72^\circ 49' = 72^\circ + (49' / 60') = \mathbf{72.82^\circ}$ (as $60' = 1^\circ$)

$$LST \text{ (in minutes)} = \mathbf{900 - 4 \times (82.5^\circ - 72.82^\circ) + (-0.24)}$$

$$LST \text{ (in minutes)} = \mathbf{861.04 \text{ min} = 14.35 \text{ hr} = 14 \text{ hrs } 21 \text{ min}}$$

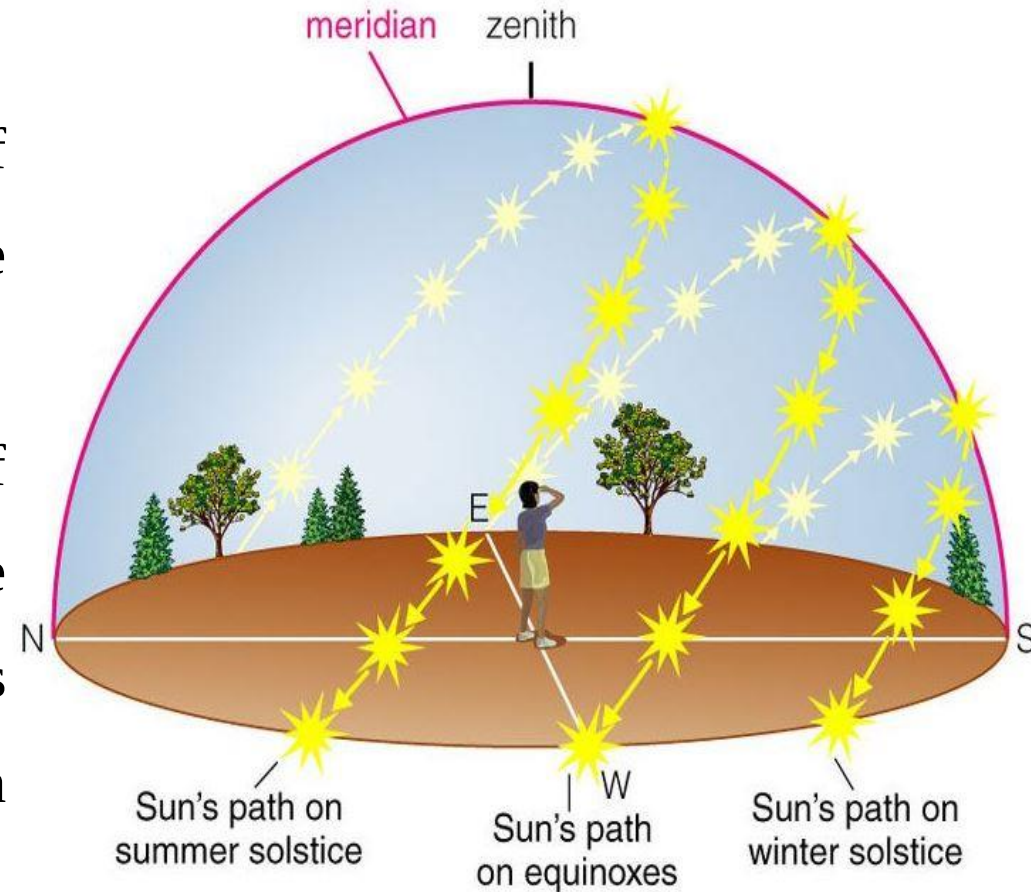
➤ **Hour angle, $\omega = 15 (LST - 12)$ in degree, LST in hours.**

$$\text{Hour angle, } \omega = 15 (\mathbf{14.35 - 12})$$

$$\text{Hour angle, } \omega = \mathbf{35.25^\circ}$$

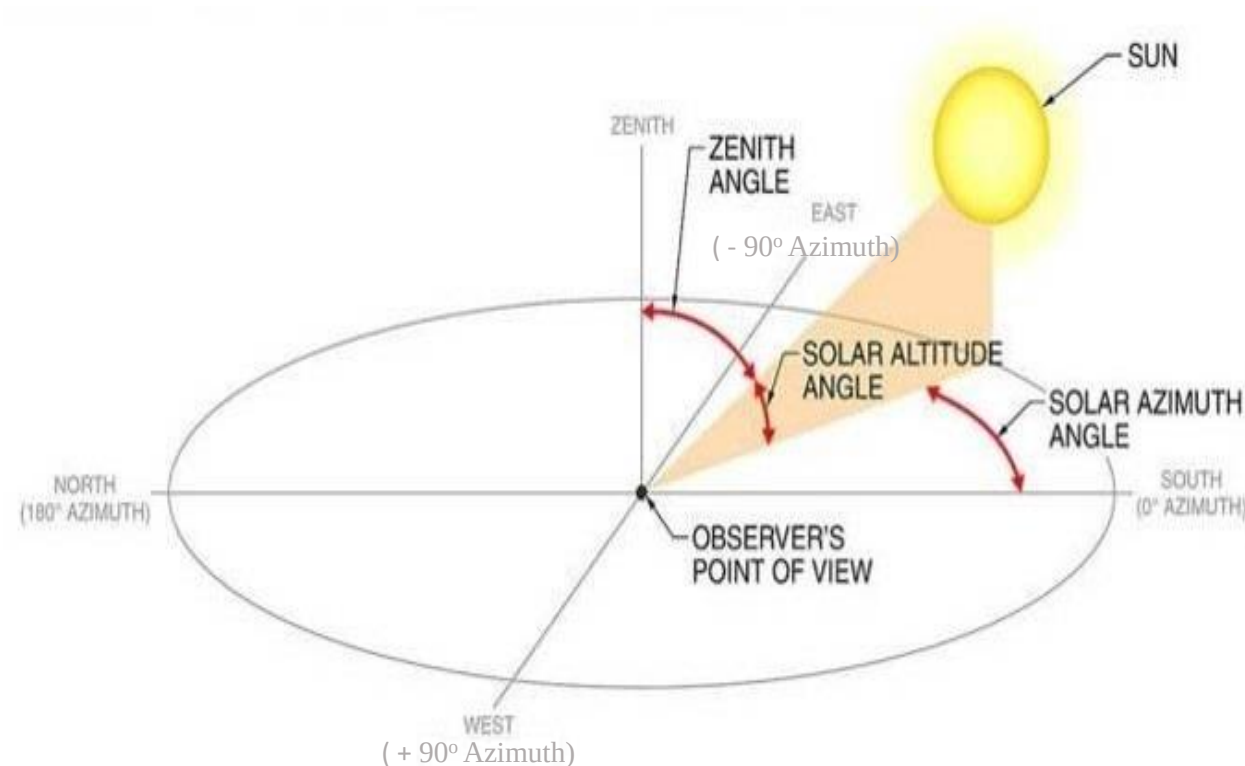
Apparent Motion of Sun

- For an observer on earth, on given day, the sun rises in the east, moves in a plane tilted at an angle of $(90^\circ - \Phi)$ with horizontal, and finally sets in the west.
- For Equator, $\Phi = 0^\circ$, sun moves in a plane normal to the horizontal *i.e.* at 90° with horizontal.
- Thus, the apparent plane in which the sun moves, intersect the horizontal plane in a line pointing east-west.
- However, because of declination angle, east-west line of intersection does not coincide with the east-west line passing through the observer.
- For the location in northern hemisphere, the line of intersection is to be south in winter (declination angle negative); to the north in summer (declination angle is positive); and coincide with east-west line passing through observer on equinox days (declination angle is zero).



Derived Solar angles for Horizontal Surface

- **Zenith angle (θ_z):** is the angle between the sun's ray (beam radiation) and line normal to a horizontal plane through the observer's location.
 - Zenith angle is known as incidence angle of solar radiation on the horizontal surface.
- **Solar Altitude Angle (α_s) :** is a vertical angle between the projection of sun's ray (beam radiation) on horizontal plane and the direction of sun's ray.
- **Solar Azimuth Angle (γ_s) :** is the horizontal angle measured from south to the projection of sun's ray (beam radiation) on horizontal plane.
- Solar azimuth angle is “**−Ve**” when projection of sun's ray is **east of south** and “**+Ve**” when projection of sun's ray is **west of south**.



Cont...

- **Zenith angle (θ_z):**

$$\cos \theta_z = \cos \Phi \cos \delta \cos \omega + \sin \Phi \sin \delta$$

- **Solar Altitude Angle (α_s) :**

$$\alpha_s = \frac{\pi}{2} - \theta_z$$

- **Solar Azimuth Angle (γ_s) :**

$$\cos \gamma_s = \frac{(\cos \theta_z \sin \Phi - \sin \delta)}{(\sin \theta_z \cos \Phi)}$$

- The attention must be given to correct sign for latitude, declination angle and hour angle.
- Latitude (Φ) for the northern hemisphere is “+**Ve**”, whereas “–**Ve**” for southern hemisphere.
- Declination angle (δ) is “+**Ve**” for summer solastic, whereas “–**Ve**” for winter solastic.
- Hour angle (ω) is “– **Ve**” for forenoon period, whereas “+**Ve**” for afternoon period.
- The beauty is sign conventions of declination angle (δ) and hour angle (ω) are taken care by its equations.

Sun rise, Sun Set and Day length for Horizontal Surface

- An hour angle corresponds to sun rise and sun set (ω_s) for the horizontal surface can be obtained by substituting the zenith angle (θ_z) equal to 90° in equation of zenith angle.

$$\cos\theta_z = \cos\Phi \cos\omega \cos\delta + \sin\Phi \sin\delta$$

$$\cos 90^\circ = 0 = \cos\Phi \cos\omega_s \cos\delta + \sin\Phi \sin\delta$$

$$\cos\omega_s = -\frac{\sin\Phi \sin\delta}{\cos\Phi \cos\delta}$$

- **Sun rise and Sun set hour angle (ω_s , in degree)** is,

$$\omega_s = \cos^{-1}(-\tan\Phi \tan\delta)$$

The equation provides two value of ω_s , “– Ve” for sun rise whereas “+Ve” for sun set.

- **The Day length (T_d , in hours)** is,

$$T_d = \frac{2}{15} \omega_s$$

Problem-2

- For the horizontal surface, determine the zenith angle, solar altitude angle, solar azimuth angle and air mass at 3:00 pm (IST) on June 15 for location (**18° 54' N, 72° 49' E**). Also, calculate the sun rise hour angle or time and day length.

➤ **Solution:**

$$\cos\theta_z = \cos\Phi \cos\delta \cos\omega + \sin\Phi \sin\delta$$

- From problem-1, declination angle, $\delta = 23.31^\circ$, hour angle, $\omega = 35.25^\circ$
- $\Phi = 18^\circ 54' \text{ N} = 18^\circ + (54' / 60') = 18.90^\circ$ (as $60' = 1^\circ$)

$$\cos\theta_z = \cos 18.90^\circ \cos 23.31^\circ \cos 35.25^\circ + \sin 18.90^\circ \sin 23.31^\circ = 0.8377$$

$$\text{Zenith angle, } \theta_z = \cos^{-1} 0.8377 = 33.10^\circ$$

$$\text{Solar altitude angle, } \alpha_s = \frac{\pi}{2} - \theta_z = \frac{\pi}{2} - 33.10^\circ = 56.90^\circ$$

Cont...

$$\cos \gamma_s = \frac{(\cos \theta_z \sin \Phi - \sin \delta)}{(\sin \theta_z \cos \Phi)} = \frac{(\cos 33.10^\circ \sin 18.90^\circ - \sin 23.31^\circ)}{(\sin 33.10^\circ \cos 18.90^\circ)} = -0.2407$$

Solar Azimuth angle, $\gamma_s = \cos^{-1}(-0.2407) = 103.92^\circ$ (west of south)

➤ *Air mass, $m = \sec \theta_z$*

$$m = \sec 33.10^\circ = 1.1937$$

➤ *Sun rise hour angle, $\omega_s = \cos^{-1}(-\tan \Phi \tan \delta)$*

$$\omega_s = \cos^{-1}(-\tan 18.90^\circ \tan 23.31^\circ) = \cos^{-1}(-0.1475) = 98.48^\circ$$

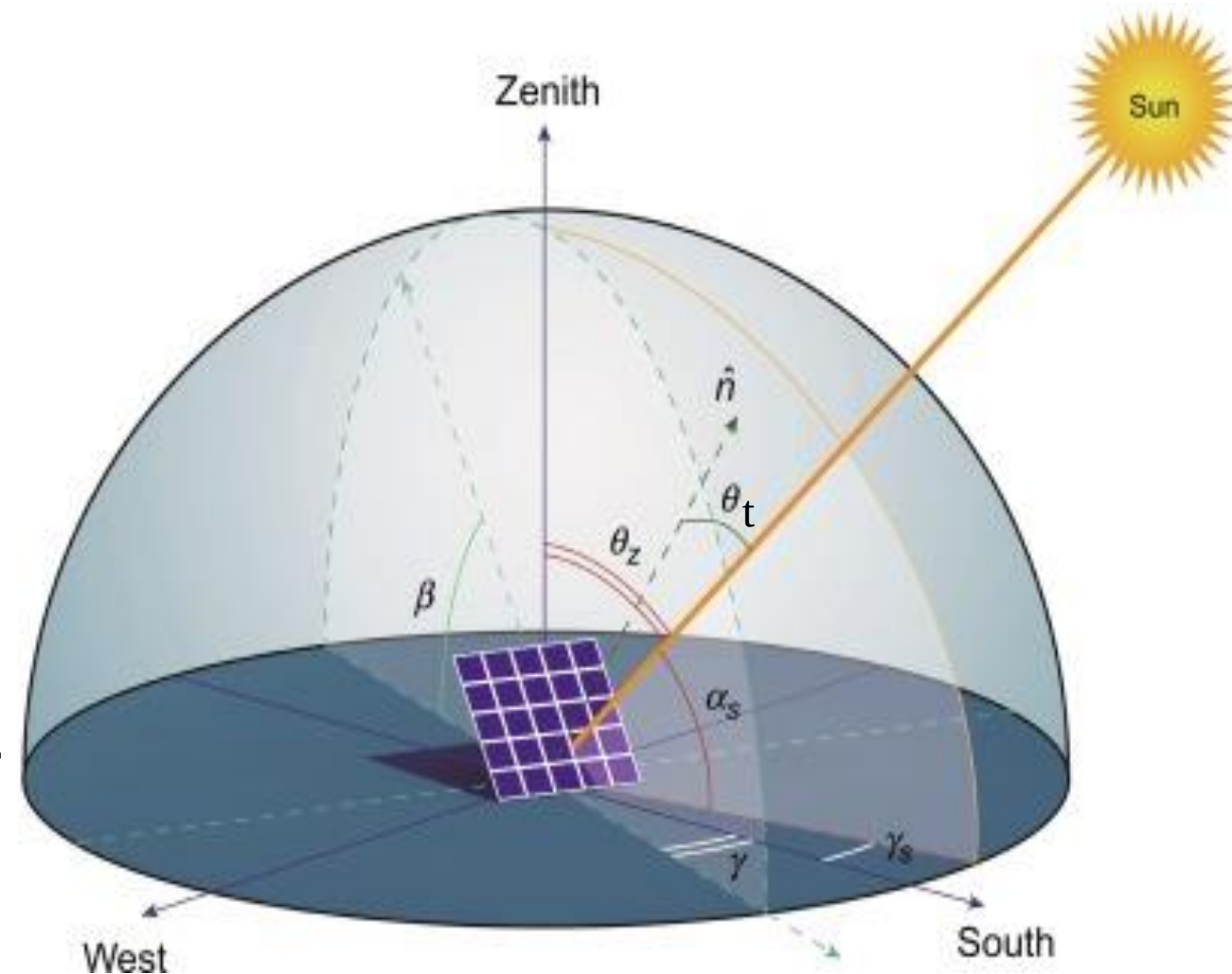
$$\text{Sun rise hour or time} = 12 - \frac{98.48^\circ}{15} = 5.4347 \text{ hr} = 5 \text{ hr } 26 \text{ min } 5 \text{ sec (prior to solar noon)}$$

➤ *Day length, $T_d = \frac{2}{15} \omega_s$*

$$\text{Day length, } T_d = \frac{2}{15} \times 98.48^\circ = 13.1307 \text{ hrs} = 13 \text{ hrs } 7 \text{ min } 50 \text{ sec}$$

Derived Solar angles for Tilted (Non-horizontal) Surface

- **Slope (β)**: is the angle between tilted / inclined surface and horizontal plane.
- **Incident angle (θ_t)** : is the angle between the normal (perpendicular) to the tilted / inclined surface and the sun ray's (Beam radiation).
- **Surface Azimuth Angle (γ)** : is the angle made in horizontal plane between the line due south and the projection of normal to the tilted / inclined surface on horizontal plane.
- Surface azimuth angle is “**− Ve**” when projection of normal to tilted surface is **east of south** and “**+Ve**” when projection of normal to tilted surface is **west of south**.



Cont...

- From the geometry of sphere,
- **Incident angle (θ_t)** for the tilted / inclined surface:

$$\begin{aligned} \cos\theta_t &= \cos\Phi \cos\delta \cos\omega \cos\beta + \sin\Phi \sin\delta \cos\beta + \cos\delta \cos\omega \cos\gamma \sin\Phi \sin\beta \\ &+ \cos\delta \sin\omega \sin\gamma \sin\beta - \cos\Phi \cos\gamma \sin\delta \sin\beta \end{aligned}$$

- For the horizontal surface, $\beta = 0^\circ$,

$$\begin{aligned} \cos\theta_t &= \cos\Phi \cos\delta \cos\omega \cos 0^\circ + \sin\Phi \sin\delta \cos 0^\circ + \cos\delta \cos\omega \cos\gamma \sin\Phi \sin 0^\circ \\ &+ \cos\delta \sin\omega \sin\gamma \sin 0^\circ - \cos\Phi \cos\gamma \sin\delta \sin 0^\circ \end{aligned}$$

$$\cos\theta_t = \cos\Phi \cos\delta \cos\omega + \sin\Phi \sin\delta = \cos\theta_z$$

Cont...

- For the vertical surface, $\beta = 90^\circ$,

$$\begin{aligned} \cos\theta_t &= \cos\Phi \cos\delta \cos\omega \cos 90^\circ + \sin\Phi \sin\delta \cos 90^\circ + \cos\delta \cos\omega \cos\gamma \sin\Phi \sin 90^\circ \\ &\quad + \cos\delta \sin\omega \sin\gamma \sin 90^\circ - \cos\Phi \cos\gamma \sin\delta \sin 90^\circ \\ \cos\theta_t &= \cos\delta \cos\omega \cos\gamma \sin\Phi + \cos\delta \sin\omega \sin\gamma - \cos\Phi \cos\gamma \sin\delta \end{aligned}$$

- For the tilted / inclined surface facing due south, $\gamma = 0^\circ$,

$$\begin{aligned} \cos\theta_t &= \cos\Phi \cos\delta \cos\omega \cos\beta + \sin\Phi \sin\delta \cos\beta + \cos\delta \cos\omega \cos 0^\circ \sin\Phi \sin\beta + \cos\delta \sin\omega \sin 0^\circ \sin\beta \\ &\quad - \cos\Phi \cos 0^\circ \sin\delta \sin\beta \\ \cos\theta_t &= \cos\Phi \cos\delta \cos\omega \cos\beta + \sin\Phi \sin\delta \cos\beta + \cos\delta \cos\omega \sin\Phi \sin\beta - \cos\Phi \sin\delta \sin\beta \\ \cos\theta_t &= \cos\delta \cos\omega (\cos\Phi \cos\beta + \sin\Phi \sin\beta) + \sin\delta (\sin\Phi \cos\beta - \cos\Phi \sin\beta) \\ \cos\theta_t &= \cos\delta \cos\omega \cos(\Phi - \beta) + \sin\delta \sin(\Phi - \beta) \end{aligned}$$

Sun rise, Sun set and Day length for Tilted Surface facing due South

- An hour angle corresponds to sun rise and sun set (ω_{st}) for the tilted / inclined surface facing due south can be obtained by substituting the angle of incidence (θ_t) equal to 90° in equation of angle of incidence for the tilted / inclined surface facing due south.

$$\cos\theta_t = \cos(\Phi - \beta) \cos\omega \cos\delta + \sin(\Phi - \beta) \sin\delta$$

$$\cos 90^\circ = 0 = \cos(\Phi - \beta) \cos\omega_{st} \cos\delta + \sin(\Phi - \beta) \sin\delta$$

$$\cos\omega_{st} = -\frac{\sin(\Phi - \beta) \sin\delta}{\cos(\Phi - \beta) \cos\delta}$$

- **Sun rise and Sun set hour angle (ω_{st} , in degree)** is,

$$\omega_{st} = \cos^{-1}(-\tan(\Phi - \beta) \tan\delta)$$

The equation provides two value of ω_s , “– Ve” for sun rise whereas “+Ve” for sun set.

- **The Day length (T_{dt} , in hours)** is,

$$T_{dt} = \frac{2}{15} \omega_{st}$$

Problem-3

- Determine the angle of incident on the surface tilted at an angle of 15° with horizontal and facing due south at 3:00 pm (IST) on June 15 for location (**$18^\circ 54'$ N, $72^\circ 49'$ E**). Also, calculate the sun rise hour or time and day length for tilted surface.

➤ **Solution:**

$$\cos\theta_t = \cos(\Phi - \beta) \cos\delta \cos\omega + \sin(\Phi - \beta) \sin\delta$$

- From problem-1, declination angle, $\delta = 23.31^\circ$, hour angle, $\omega = 35.25^\circ$
- $\Phi = 18^\circ 54' \text{ N} = 18^\circ + (54' / 60') = 18.90^\circ$ (as $60' = 1^\circ$)
- $\beta = 15^\circ$

$$\cos\theta_t = \cos(18.90^\circ - 15^\circ) \cos 23.31^\circ \cos 35.25^\circ + \sin(18.90^\circ - 15^\circ) \sin 23.31^\circ$$

$$\cos\theta_t = 0.7751$$

$$\text{Incident angle, } \theta_t = \cos^{-1} 0.7751 = 39.18^\circ$$

Cont...

➤ *Sun rise hour angle, $\omega_{St} = \cos^{-1}(-\tan(\Phi - \beta)\tan\delta)$*

$$\omega_{St} = \cos^{-1}(-\tan(18.90^\circ - 15^\circ) \tan 23.31^\circ)$$

$$\omega_{St} = \cos^{-1}(-0.0294) = 91.68^\circ$$

Sun rise hour or time = $12 - \frac{91.68^\circ}{15} = 5.888 \text{ hr} = 5 \text{ hr } 53 \text{ min } 17 \text{ sec}$ (Prior to solar noon)

➤ *Day length, $T_{dt} = \frac{2}{15} \omega_{St}$*

$$\text{Day length, } T_{dt} = \frac{2}{15} \times 91.68^\circ = 12.224 \text{ hrs} = 12 \text{ hrs } 13 \text{ min } 26 \text{ sec}$$

Problem-4

- A surface is tilted at an angle of 40° and faces 5° east of south at location (**21.18° N, 72.83° E**). Determine an angle of incident the at (i) 12:00 pm (IST), (ii) 12:00 pm (LAT) on 23rd December.

➤ **Solution:**

➤ **For (i) 12:00 pm IST**

Declination angle (in degree), $\delta = 23.45 \sin \left(\frac{360 \times (284 + n)}{365} \right)$

▪ $n = 31 + 28 + 31 + 30 + 31 + 30 + 31 + 31 + 30 + 31 + 30 + 23 = 357$ days

$$\delta = 23.45 \sin \left(\frac{360 \times (284 + 357)}{365} \right) = -23.432^\circ$$

$$\text{LST (in minutes)} = \text{ST} - 4 \times (\text{L}_{\text{ST}} - \text{L}_{\text{Local}}) + \text{E.O.T.}$$

▪ $\text{E.O.T.} = 9.87 \sin 2B - 7.53 \cos B - 1.5 \sin B$

Cont...

- $B = \frac{360 (n-81)}{364} = \frac{360 (357-81)}{364} = \mathbf{272.967^\circ}$
- $E.O.T. = 9.87 \sin(2 \times \mathbf{272.967^\circ}) - 7.53 \cos(\mathbf{272.967^\circ}) - 1.5 \sin(\mathbf{272.967^\circ}) = \mathbf{0.088 \text{ min}}$
- $ST = \mathbf{12 \text{ pm IST}} = 12 \times 60 = \mathbf{720 \text{ min}}$
- $L_{ST} = \mathbf{82.5^\circ}$
- $L_{Local} = \mathbf{72.83^\circ}$

$$\mathbf{LST \text{ (in minutes)} = 720 - 4 \times (82.5^\circ - 72.83^\circ) + 0.088}$$

$$\mathbf{LST \text{ (in minutes)} = 681.408 \text{ min} = 11.357 \text{ hr} = 11 \text{ hr } 21 \text{ min } 25 \text{ sec}}$$

➤ **Hour angle, $\omega = 15 (LST - 12)$ in degree,**

$$\text{Hour angle, } \omega = 15 (\mathbf{11.357} - 12)$$

$$\text{Hour angle, } \omega = \mathbf{-9.645^\circ}$$

Cont...

➤ **Incident angle (θ_t)** for the tilted / inclined surface:

$$\cos\theta_t$$

$$= \cos\Phi \cos\delta \cos\omega \cos\beta + \sin\Phi \sin\delta \cos\beta + \cos\delta \cos\omega \cos\gamma \sin\Phi \sin\beta \\ + \cos\delta \sin\omega \sin\gamma \sin\beta - \cos\Phi \cos\gamma \sin\delta \sin\beta$$

▪ $\gamma = 5^\circ$ east of south = -5°

$$\cos\theta_t$$

$$= \cos 21.18^\circ \cos(-23.432^\circ) \cos(-9.645^\circ) \cos 40^\circ + \sin 21.18^\circ \sin(-23.432^\circ) \cos 40^\circ \\ + \cos(-23.432^\circ) \cos(-9.645^\circ) \cos(-5^\circ) \sin 21.18^\circ \sin 40^\circ \\ + \cos(-23.432^\circ) \sin(-9.645^\circ) \sin(-5^\circ) \sin 40^\circ - \cos 21.18^\circ \cos(-5^\circ) \sin(-23.432^\circ) \sin 40^\circ$$

$$\cos\theta_t = 0.991$$

$$\text{Angle of incident, } \theta_t = \cos^{-1} 0.991 = 7.523^\circ$$

Cont...

➤ **For (ii) 12:00 pm LAT**

■ Hour angle, $\omega = 15 (12 - 12) = 0^\circ$

➤ **Incident angle (θ_t) for the tilted / inclined surface:**

$$\cos\theta_t$$

$$= \cos\Phi \cos\delta \cos\omega \cos\beta + \sin\Phi \sin\delta \cos\beta + \cos\delta \cos\omega \cos\gamma \sin\Phi \sin\beta \\ + \cos\delta \sin\omega \sin\gamma \sin\beta - \cos\Phi \cos\gamma \sin\delta \sin\beta$$

$$\cos\theta_t$$

$$= \cos 21.18^\circ \cos(-23.432^\circ) \cos(0^\circ) \cos 40^\circ + \sin 21.18^\circ \sin(-23.432^\circ) \cos 40^\circ \\ + \cos(-23.432^\circ) \cos(0^\circ) \cos(-5^\circ) \sin 21.18^\circ \sin 40^\circ + \cos(-23.432^\circ) \sin(0^\circ) \sin(-5^\circ) \sin 40^\circ \\ - \cos 21.18^\circ \cos(-5^\circ) \sin(-23.432^\circ) \sin 40^\circ$$

$$\cos\theta_t = 0.9950$$

$$\text{Angle of incident, } \theta_t = \cos^{-1} 0.9950 = 5.71^\circ$$

Reference

- **Solar Energy: Principles of Thermal Collection and Storage by S. P. Sukhatme and J. K. Nayak, 3rd Edition, Tata McGraw Hill Education Private limited, New Delhi.**